Class05: Data Viz with ggplot2

Yassaman Fakhari (PID A18541145)

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Background

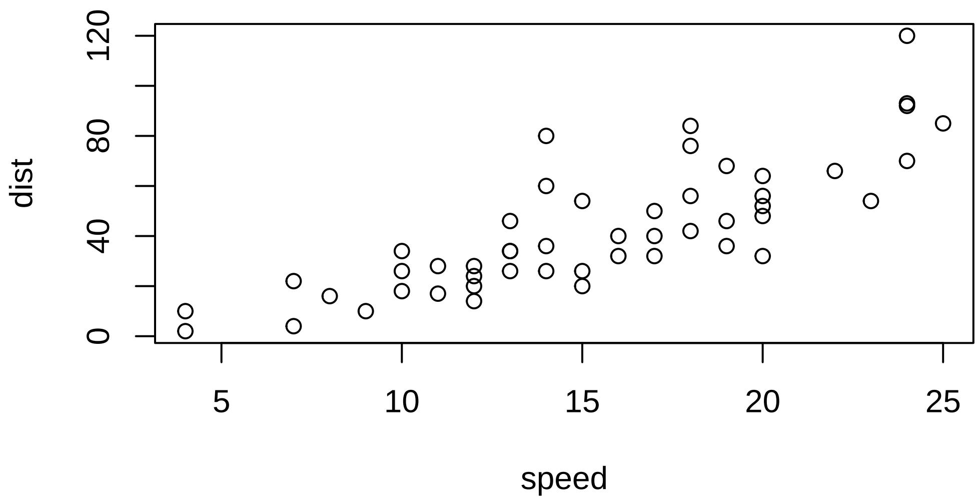
There are many graphics systems in R for making plots and figures. These include so-called *"base R" graphics* like the 'plot()' function and add on packages like **ggplot2**

Let's compare how we make a simple figure with these two systems: We can use the in-built "cars" dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```



Before I can use ggplot2 I need to install it on my computer. To do this we can use the function 'install.packages(ggplot2)'

Once installed we need to load up the package into our R brain:

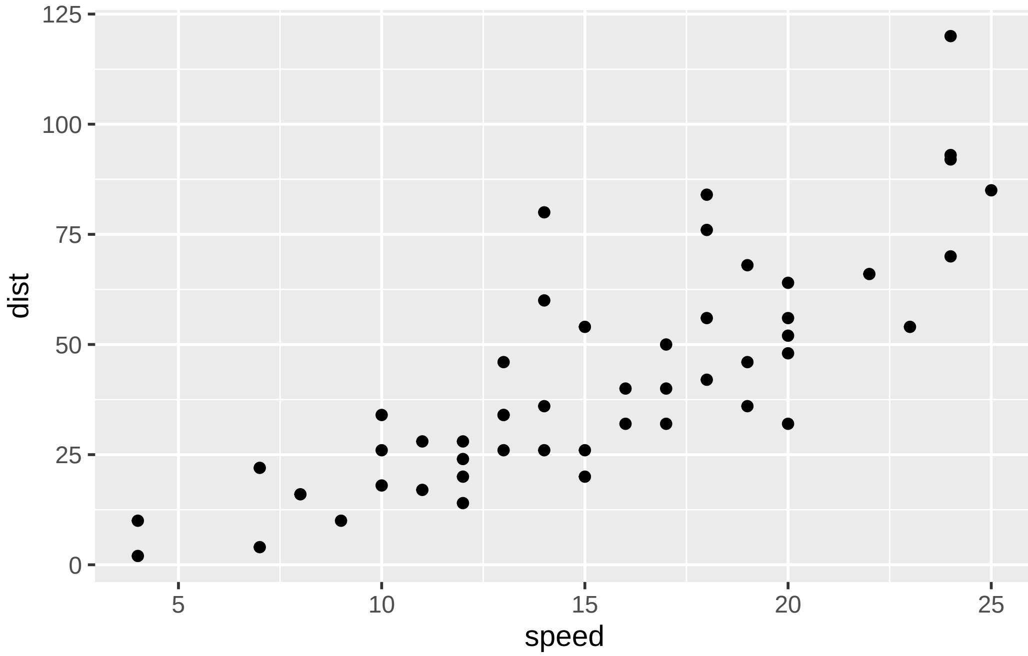
```
library(ggplot2)
```

The main function in the **ggplot** package is called ggplot()

```
ggplot(cars)
```

every ggplot has at least 3 layers: - the **data** (a data.frame of the stuff we want to plot) - the **aes**thetics (how the data maps on the plot) - the **geom** layer (how you want the plot drawn, ex. points, lines, etc.)

```
ggplot(cars, aes(x=speed, y=dist)) +
  geom_point()
```

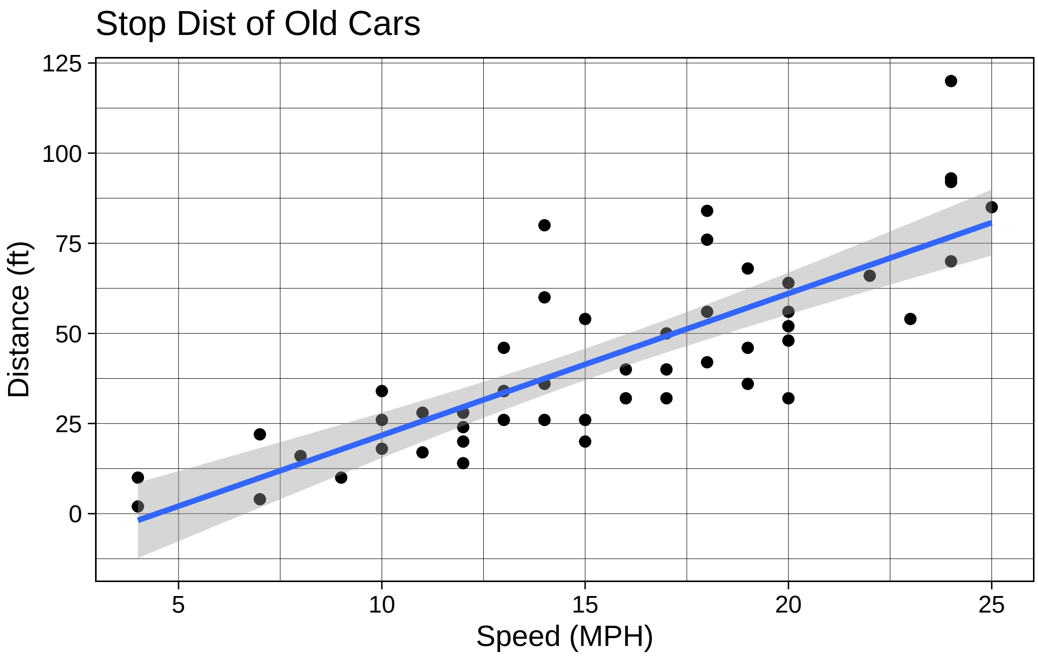


Add Some Cutsom Features

Let's add a trend line that shows the relationsip between speed and distance.

```
ggplot(cars, aes(x=speed, y=dist)) +
  geom_point() +
  geom_smooth(method = "lm") +
  theme_linedraw() +
  labs(title="Stop Dist of Old Cars",
       x="Speed (MPH)",
       y= "Distance (ft)")
```

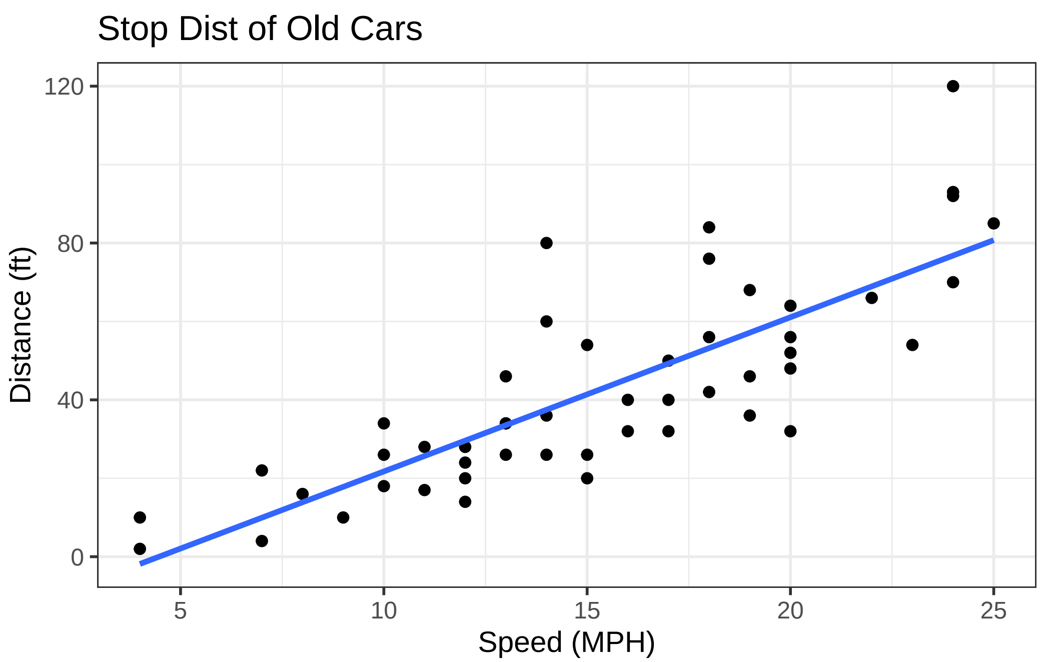
```
`geom_smooth()` using formula = 'y ~ x'
```



Q. Can you make the 'geom_smooth()' function produces a linear straight line fit to the data and turn off the 'grey' error region

```
ggplot(cars, aes(x=speed, y=dist)) +
  geom_point() +
  geom_smooth(method = "lm", se=FALSE) +
  theme_bw() +
  labs(title="Stop Dist of Old Cars",
       x="Speed (MPH)",
       y= "Distance (ft)")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Gene Expression Figure

```r
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```r
sum(genes$State == "up")
```

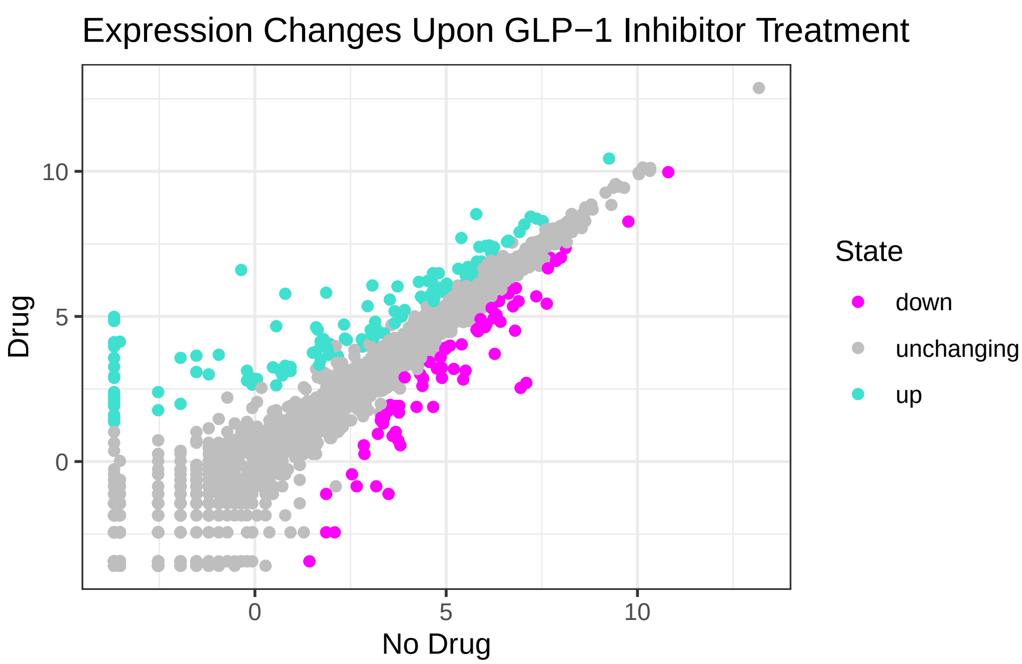
```
[1] 127
```

```r
table(genes$State)
```

```
      down unchanging         up 
        72       4997        127 
```

My first plot attempt

```r
ggplot(genes) +
  aes(x=Condition1, y=Condition2 ,col=State) +
  geom_point() +
  scale_color_manual(values=c("magenta", "gray", "turquoise")) +
  theme_bw() +
  labs(x="No Drug", y="Drug",
       title="Expression Changes Upon GLP-1 Inhibitor Treatment")
```



Going Further

Here we read the famous gapmider dataset:

```r
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many entries (ex. rows) are in this dataset?

```r
nrow(gapminder)
```

```
[1] 1704
```

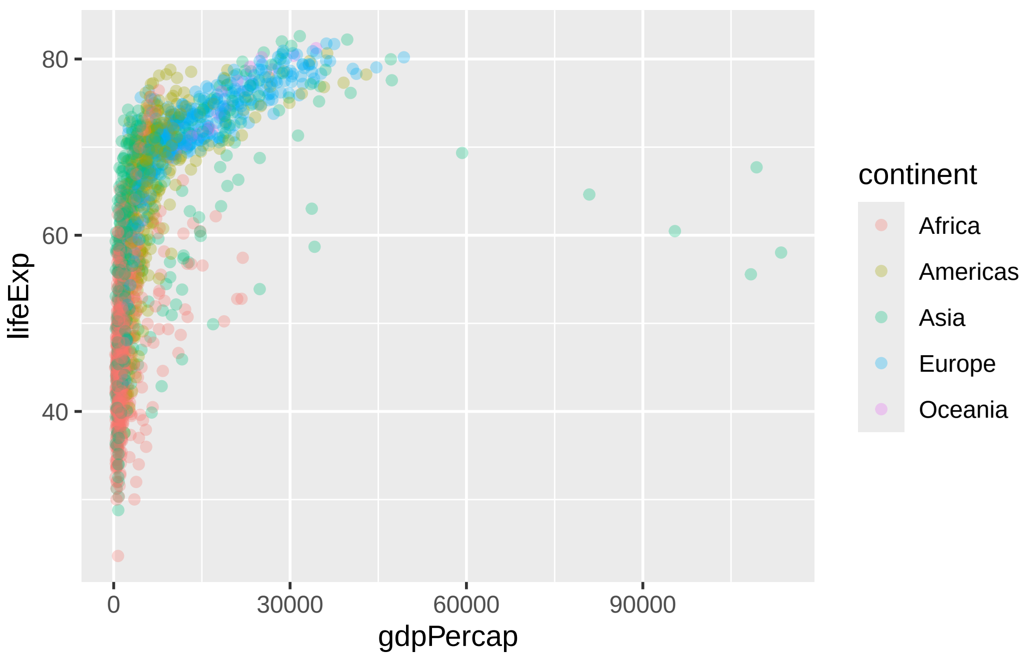
Q. How many different "country" enteries are in this dataset?

```r
length(unique(gapminder$country))
```

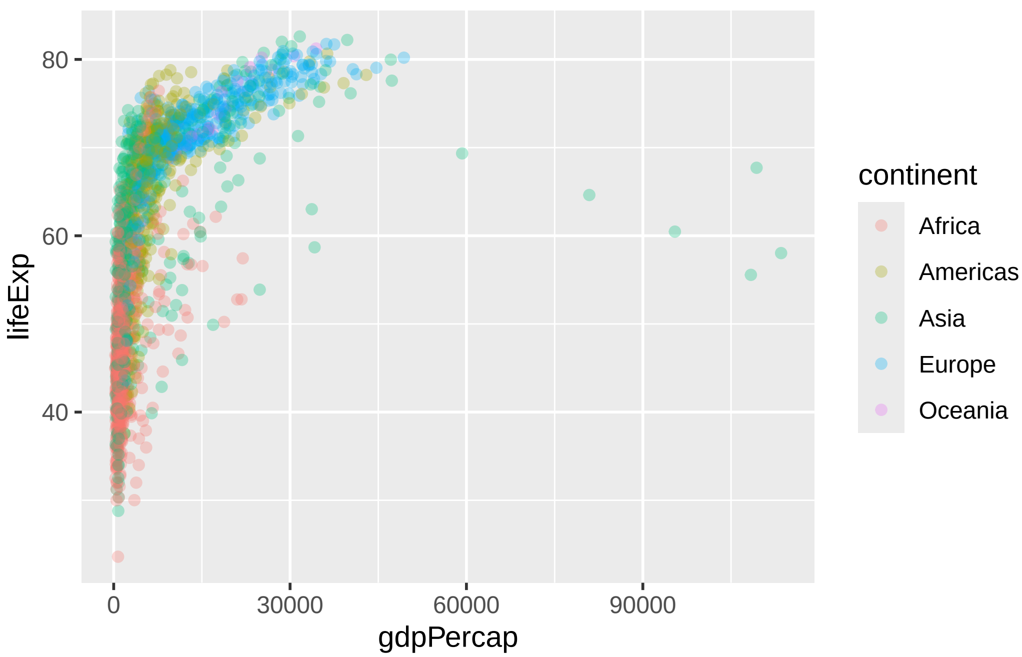
```
[1] 142
```

Let's make our first plot of the entire dataset: Plot of "gdpPercap" vs "lifeExp" colored by "continent"

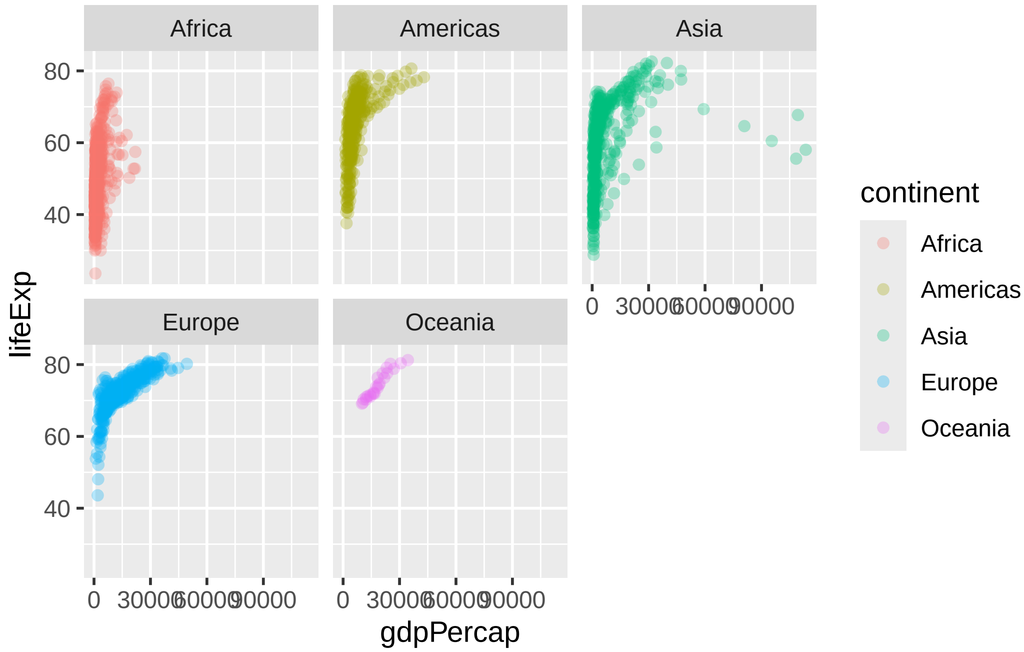
```r
p <- ggplot(gapminder) +
  aes(gdpPercap, lifeExp, col=continent) +
  geom_point(alpha=0.3)
p
```


80
60
40
lifeExp
0
30000
60000
90000
gdpPercap
continent
Africa
Americas
Asia
Europe
Oceania

```
p
```


80
60
40
lifeExp
0
30000
60000
90000
gdpPercap
continent
Africa
Americas
Asia
Europe
Oceania

```
p +
  facet_wrap(~continent)
```



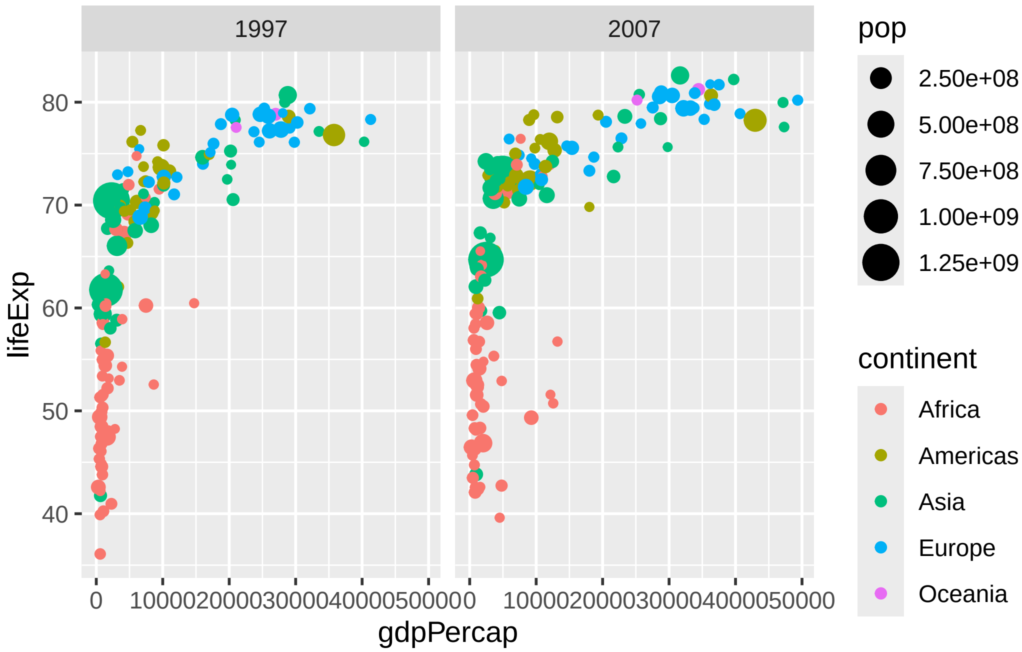
Make a plot for years 1977 and 2007 only (not all the years in the dataset).

> Q. First use the **dplyr** package and the 'filter()' function from that package to extract the rows from the year 2007

```
library(dplyr)
```

```
g07 <- filter(gapminder, year == 2007)
g77 <- filter(gapminder, year == 1977)
g <- filter(gapminder, year == 2007 | year == 1997)
```

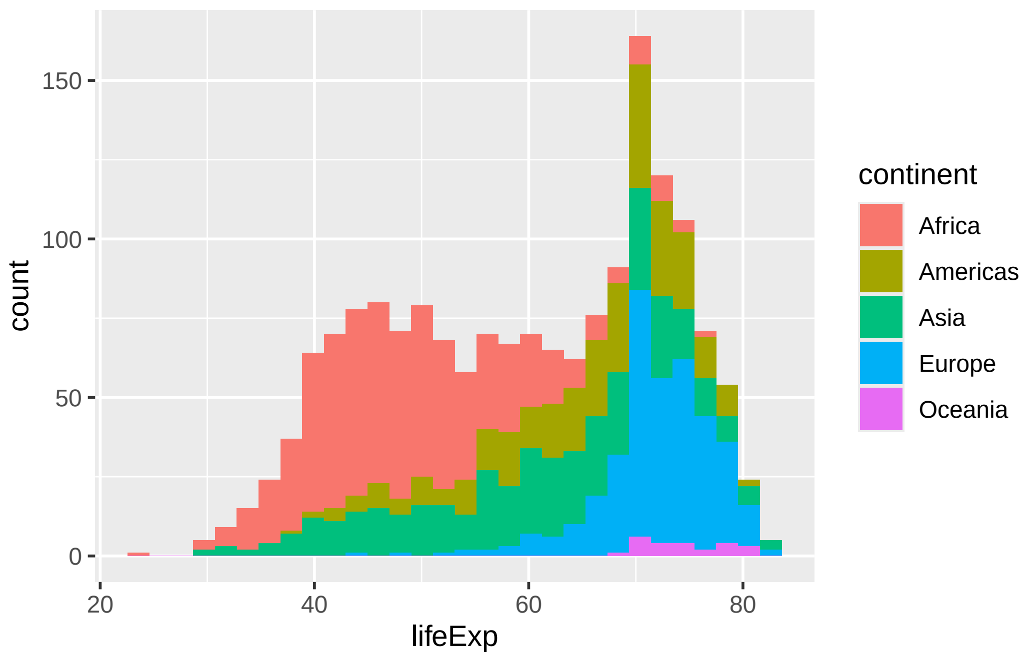
```
ggplot(g) +
  aes(gdpPercap, lifeExp, col=continent, size=pop) +
  geom_point() +
  facet_wrap(~year)
```



Q. Make a histogram of lifeExp colored by continent (try using 'fill=continent' or 'col=continent')

```
ggplot(gapminder) +
  aes(lifeExp, fill=continent) +
  geom_histogram()
```

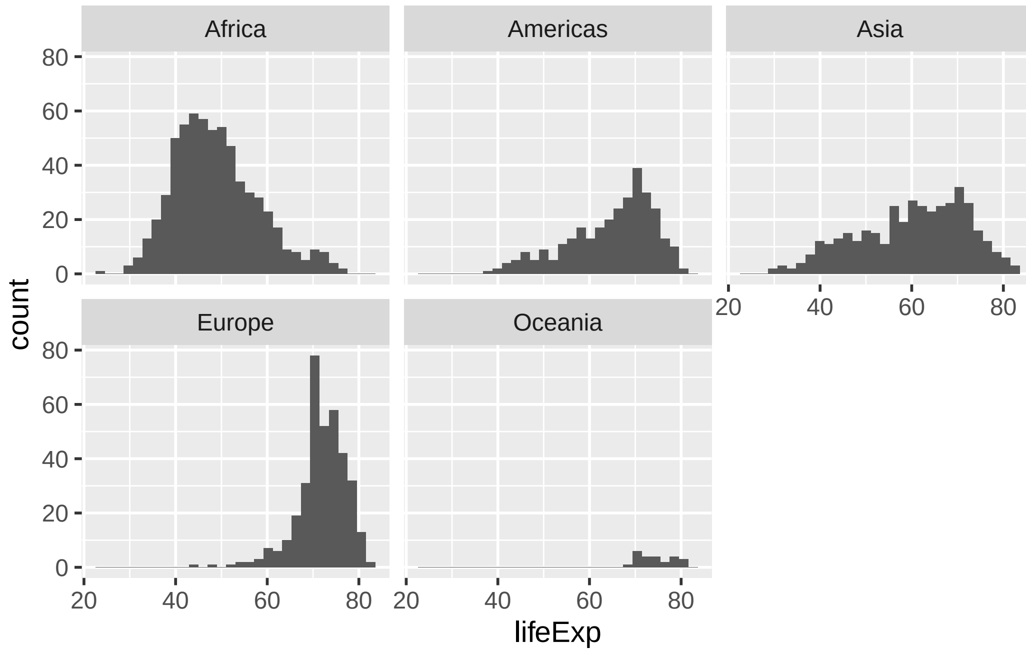
```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Q. Make a histogram of lifeExp faceted by continent

```
ggplot(gapminder) +
  aes(lifeExp) +
  geom_histogram() +
  facet_wrap(~continent)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```


Africa
Americas
Asia
Europe
Oceania
count
lifeExp
0
20
40
60
80